

VARIABLE UNIDIMENSIONAL Y BIDIMENSIONAL

$$f_i = \frac{n_i}{n} ; \bar{x} = \frac{\sum_{i=1}^n x_i}{n} \quad \bar{x} = \frac{\sum_{i=1}^k x_i * n_i}{n} ; Q_1 = \frac{n}{4} ; Q_2 = Me ; Q_3 = \frac{3n}{4} ; P = \frac{k*n}{100} ; C.V = \frac{s}{\bar{x}}$$

Mediana:

$$1) n/2 ; \quad 2) Si N_{i-1} < \frac{n}{2} < N_i \rightarrow Me = x_i$$
$$2) Si N_{i-1} = \frac{n}{2} < N_i \rightarrow Me = \frac{x_{i-1} + x_i}{2} ; 2) Si N_{i-1} \leq \frac{n}{2} < N_i \rightarrow [L_{i-1}, L_i) ; 3) Me$$
$$= L_{i-1} + \frac{n/2 - N_{i-1}}{n_i} * c_i$$

Moda:

1) Se observa el mayor n_i (o el mayor $k_i = n_i/c_i$) $\rightarrow$ Md está en el intervalo $[L_{i-1}, L_i)$

$$2) \text{ mismo } c_i : Md = L_{i-1} + \frac{n_i - n_{i-1}}{(n_i - n_{i-1}) + (n_i - n_{i+1})} * c_i$$

$$2) \text{ distinto } c_i : Md = L_{i-1} + \frac{k_i - k_{i-1}}{(k_i - k_{i-1}) + (k_i - k_{i+1})} * c_i$$

Percentiles, deciles y cuartiles: $(P \cdot n/k)$

$$P = L_{i-1} + \frac{(P \cdot n/k) - N_{i-1}}{n_i} \cdot C_i$$

Rango intercuartílico (valores atípicos): $RI = Q_3 - Q_1$

$q < Q_1 - 1.5 * RI$ y $q > Q_3 + 1.5 * RI$ (Valores atípicos leves)

$q < Q_1 - 3 * RI$ y $q > Q_3 + 3 * RI$ (Valores atípicos extremo)

Varianza:

$$s^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n} = \frac{\sum_{i=1}^n x_i^2}{n} - \bar{x}^2 = \frac{\sum_{i=1}^k n_i * (x_i - \bar{x})^2}{n} = \frac{\sum_{i=1}^k n_i * x_i^2}{n} - \bar{x}^2$$

$$s = \sqrt{s^2} ; C.V = \frac{s}{\bar{x}} ; Var(X + k) = Var(X) ; Var(kX) = k^2 * Var(X) ;$$

$$Var(X + Y) = Var(X) + Var(Y) + 2(Cov XY)$$

Asimetría:

$$A_F = \frac{\sum_{i=1}^n (x_i - \bar{x})^3}{n * s^3} = \frac{\sum_{i=1}^k n_i * (x_i - \bar{x})^3}{n * s^3}$$

$$A_B = \frac{(Q_3 - Me) - (Me - Q_1)}{Q_3 - Q_1}$$

$$A_p = \frac{\bar{x} - Md}{s}$$

Coeficiente de apuntamiento:

$$g = \frac{\sum_{i=1}^k n_i \cdot (x_i - \bar{x})^4}{n \cdot s^4}$$

Covarianza:

$$s_{xy} = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x}) * (y_i - \bar{y}) = \frac{1}{n} \sum_{i=1}^n x_i y_i - \bar{x} \bar{y} = \frac{1}{n} \sum_{i=1}^k \sum_{j=1}^m n_{ij} * x_i * y_j - \bar{x} \bar{y}$$

Regresión lineal:

$$y/x \rightarrow y - \bar{y} = \frac{s_{xy}}{s_x^2} (x_i - \bar{x})$$

$$y = \beta_0 + \beta_1 x \begin{cases} \beta_1 = \frac{s_{xy}}{s_x^2} \\ \beta_0 = \bar{y} - \beta_1 \bar{x} \end{cases}$$

Correlación de Pearson:

$$r_{xy} = \frac{s_{xy}}{s_x * s_y}$$

Coeficiente de determinación: $R^2 = r^2$

Varianza m: $V(Y) = s_y^2 \cdot (1 - R^2)$

Varianza e: $V(Y) = s_y^2 \cdot R^2$

PROBABILIDAD:

$$P(\bar{A}) = 1 - P(A)$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(B \cap C) - P(A \cap C) + P(A \cap B \cap C)$$

$$P(A - B) = P(A \cap \bar{B}) = P(A) - P(A \cap B)$$

$$P(\bar{A} \cap \bar{B}) = P(\overline{A \cup B})$$

$$P(\bar{A} \cup \bar{B}) = P(\overline{A \cap B})$$

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

$$\text{Indep: } P(A \cap B) = P(A) \cdot P(B)$$

$$\text{Dep: } P(A \cap B) = P(B|A) \cdot P(A)$$

- TPT: $P(B) = \sum_{i=1}^n P(B|A_i) \cdot P(A_i)$

- TB: $P(A_i|B) = \frac{P(B|A_i) \cdot P(A_i)}{\sum_{i=1}^n P(B|A_i) \cdot P(A_i)} = \frac{P(A_i \cap B)}{P(B)}$

DISTRIBUCIONES DISCRETAS:

Variables Unidimensionales discretas

Esperanza matemática, valor esperado o Media poblacional:

$$\mu = E[X] = \sum_{i=1}^n x_i * p(x_i)$$

$$E[Y] = E[aX + b] = a * E[X] + b$$

Varianza poblacional:

$$\sigma^2 = \sum_{i=1}^n (x_i - \mu)^2 * p(x_i) = \sum_{i=1}^n x_i^2 * p(x_i) - \mu^2 = E(X^2) - (E[X])^2$$

Desviación poblacional:

$$\sigma = +\sqrt{\sigma^2}$$

$$Var[Y] = Var[aX + b] = a^2 * Var[X]$$

Distribución Bernoulli:

$$X = BE(p)$$

$$P(A)(\text{éxito}) = p, \quad P$$

$$(\overline{A})(\text{fracaso}) = q$$

$$\mu = n * p \mid \sigma^2 = n * p * q \mid \sigma = +\sqrt{n * p * q}$$

Distribución Binomial:

$$X = BIN(n, p)$$

$$p(X = k) = \binom{n}{k} p^k q^{n-k}$$

$$\mu = n * p \mid \sigma^2 = n * p * q \mid \sigma = +\sqrt{n * p * q}$$

• $n > 50$ y $np < 5$ o $n > 30$ y $p \leq 0,1$ $BIN(n, p) \rightarrow P(\lambda)$ ($\lambda = n * p$)

Distribución Poisson:

$$X = P(\lambda)$$

$$p(X = k) = \frac{\lambda^k}{k!} e^{-\lambda}$$

$$\mu = \lambda \mid \sigma^2 = \lambda \mid \sigma = +\sqrt{\lambda}$$

Variables Bidimensionales discretas

Función de masa de una variable bidimensional (discreta):

$$p(X = x, Y = y)$$

$$\sum \sum p(X = x_i, Y = y_j) = 1$$

$$x_i \ y_j$$

Función de distribución de una variable bidimensional:

$$F(x, y) = p(X \leq x, Y \leq y)$$

Caso X e Y discretas,

$$F(x, y) = p(X \leq x, Y \leq y) = \sum_{x_i \leq x} \sum_{y_j \leq y} p(X = x_i, Y = y_j)$$

Covarianza poblacional:

$$\sigma_{X,Y} = \sum_i^n \sum_j^m x_i * y_j * p(X = x_i, Y = y_j) - \mu_X * \mu_Y$$

Correlación:

$$r = \frac{\sigma_{X,Y}}{\sigma_X * \sigma_Y}$$

Distribuciones condicionadas:

$$p(x_i | y_j) = \frac{p(X = x_i, Y = y_j)}{p_2(X = x_i)}$$

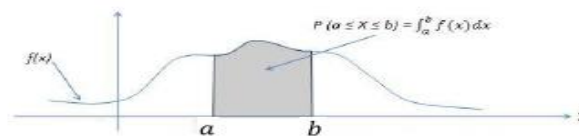
$$p(y_j | x_i) = \frac{p(X = x_i, Y = y_j)}{p_1(Y = y_j)}$$

$$\mu = \mu_1 + \mu_2 + \dots + \mu_n \mid \sigma = \sqrt{\sigma_1^2 + \sigma_2^2 + \dots + \sigma_n^2}$$

DISTRIBUCIONES CONTINUAS:

Función de densidad):

$$p(a < X < b) = \int_a^b f(x) dx$$



Función de distribución:

$$F(x) = p(X \leq x) = \int_{-\infty}^x f(t) dt$$

Esperanza matemática, Valor esperado o Media poblacional

$$\mu = E[X] = \int_{-\infty}^{+\infty} x \cdot f(x) dx$$

Varianza poblacional ($Var X = \sigma^2$)

$$\sigma^2 = Var[X] = E[(X - \mu)^2] = \int_{-\infty}^{+\infty} (x - \mu)^2 \cdot f(x) dx = \int_{-\infty}^{+\infty} x^2 \cdot f(x) dx - \mu^2$$

Distribución normal

Función de densidad:

$$f(x) = \frac{1}{\sigma \sqrt{2\pi}} \cdot e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

Función de distribución:

$$F(x) = \frac{1}{\sigma \sqrt{2\pi}} \cdot \int_{-\infty}^x e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx$$

Tipificación:

$$Z = \frac{x - \mu}{\sigma}$$

Distribución exponencial

$$f(x) = \frac{1}{\mu} e^{-\frac{1}{\mu}x}$$

$$\mu = 1/\lambda \quad | \quad \sigma^2 = 1/\lambda^2 \quad | \quad \sigma = 1/\lambda$$

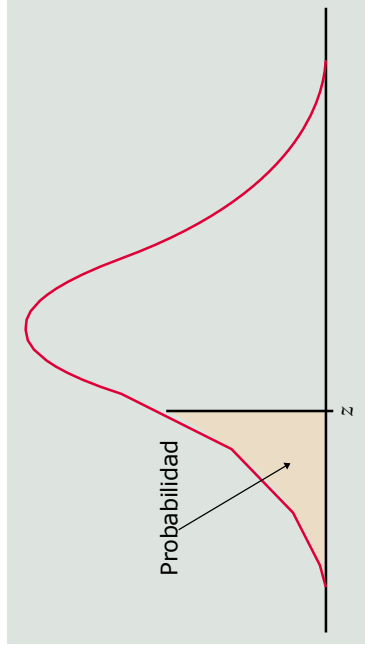


TABLA A: Probabilidades de la normal estándar

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
−3.4	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0002
−3.3	.0005	.0005	.0005	.0004	.0004	.0004	.0004	.0004	.0004	.0003
−3.2	.0007	.0007	.0006	.0006	.0006	.0006	.0006	.0005	.0005	.0005
−3.1	.0010	.0009	.0009	.0009	.0008	.0008	.0008	.0008	.0007	.0007
−3.0	.0013	.0013	.0013	.0012	.0012	.0011	.0011	.0011	.0010	.0010
−2.9	.0019	.0018	.0018	.0017	.0016	.0016	.0015	.0015	.0014	.0014
−2.8	.0026	.0025	.0024	.0023	.0023	.0022	.0021	.0021	.0020	.0019
−2.7	.0035	.0034	.0033	.0032	.0031	.0030	.0029	.0028	.0027	.0026
−2.6	.0047	.0045	.0044	.0043	.0041	.0040	.0039	.0038	.0037	.0036
−2.5	.0062	.0060	.0059	.0057	.0055	.0054	.0052	.0051	.0049	.0048
−2.4	.0082	.0080	.0078	.0075	.0073	.0071	.0069	.0068	.0066	.0064
−2.3	.0107	.0104	.0102	.0099	.0096	.0094	.0091	.0089	.0087	.0084
−2.2	.0139	.0136	.0132	.0129	.0125	.0122	.0119	.0116	.0113	.0110
−2.1	.0179	.0174	.0170	.0166	.0162	.0158	.0154	.0150	.0146	.0143
−2.0	.0228	.0222	.0217	.0212	.0207	.0202	.0197	.0192	.0188	.0183
−1.9	.0287	.0281	.0274	.0268	.0262	.0256	.0250	.0244	.0239	.0233
−1.8	.0359	.0351	.0344	.0336	.0329	.0322	.0314	.0307	.0301	.0294
−1.7	.0446	.0436	.0427	.0418	.0409	.0401	.0392	.0384	.0375	.0367
−1.6	.0548	.0537	.0526	.0516	.0505	.0495	.0485	.0475	.0465	.0455
−1.5	.0668	.0655	.0643	.0630	.0618	.0606	.0594	.0582	.0571	.0559
−1.4	.0808	.0793	.0778	.0764	.0749	.0735	.0721	.0708	.0694	.0681
−1.3	.0968	.0951	.0934	.0918	.0901	.0885	.0869	.0853	.0838	.0823
−1.2	.1151	.1131	.1112	.1093	.1075	.1056	.1038	.1020	.1003	.0985
−1.1	.1357	.1335	.1314	.1292	.1271	.1251	.1230	.1210	.1190	.1170
−1.0	.1587	.1562	.1539	.1515	.1492	.1469	.1446	.1423	.1401	.1379
−0.9	.1841	.1814	.1788	.1762	.1736	.1711	.1685	.1660	.1635	.1611
−0.8	.2119	.2090	.2061	.2033	.2005	.1977	.1949	.1922	.1894	.1867
−0.7	.2420	.2389	.2358	.2327	.2296	.2266	.2236	.2206	.2177	.2148
−0.6	.2743	.2709	.2676	.2643	.2611	.2578	.2546	.2514	.2483	.2451
−0.5	.3085	.3050	.3015	.2981	.2946	.2912	.2877	.2843	.2810	.2776
−0.4	.3446	.3409	.3372	.3336	.3300	.3264	.3228	.3192	.3156	.3121
−0.3	.3821	.3783	.3745	.3707	.3669	.3632	.3594	.3557	.3520	.3483
−0.2	.4207	.4168	.4129	.4090	.4052	.4013	.3974	.3936	.3897	.3859
−0.1	.4602	.4562	.4522	.4483	.4443	.4404	.4364	.4325	.4286	.4247
−0.0	.5000	.4960	.4920	.4880	.4840	.4801	.4761	.4721	.4681	.4641

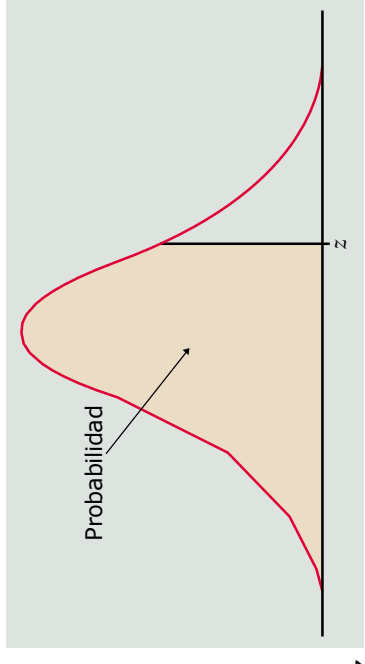


TABLA A: Probabilidades de la normal estándar (cont.)

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